PERSONAL RESUME

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Information

Name： CHEN Dan

Gender： Male

Email： danhust9986@gmail.com

Birthday： 1996-11

Phone： +65 9866 1685

Degree： PhD



Work Experience

May 2026 - Present

Black Sesame Technologies — **Architect, AI Computing**

Develop AI agents for automated chip design-space exploration and compute-kernel optimization.

Mar 2024 - Apr 2026

National University of Singapore — **Postdoctoral Research Fellow**

Advisor: Prof. Tulika Mitra, Dean, School of Computing

Education

Sep 2018 - Mar 2024

Huazhong University of Science and Technology — **Computer System Architecture (PhD)**

Advisor: Prof. Hai Jin, IEEE/CCF Fellow; Vice President of CCF; Changjiang Distinguished Professor; Recipient of the National Science Fund for Distinguished Young Scholars

Sep 2014 - Jun 2018

North China Electric Power University — **Software Engineering (Bachelor)**

Ranked first in the major; admitted directly to the PhD program at Huazhong University of Science and Technology with an entrance-examination exemption.

Research interest

My research focuses on AI infrastructure, particularly high-performance, energy-efficient computing systems and accelerator architectures developed through hardware–software co-design:

1. Efficient AI systems and accelerators for large language models, retrieval-augmented generation (RAG), and graph neural networks.

2. Near-memory and processing-in-memory architectures and computation offloading.

3. AI agents for automated chip design-space exploration and compute-kernel optimization.

Publications

Selected First-Author, Co-first-Author and Corresponding-Author Publications

* Corresponding author.

1. **MetaNMP: Leveraging Cartesian-Like Product to Accelerate HGNNs with Near-Memory Processing**. **Dan Chen,** Haiheng He, Hai Jin, Long Zheng, Yu Huang, Xinyang Shen, Xiaofei Liao. ACM/IEEE International Symposium on Computer Architecture (ISCA), 2023.

2. **HeterRAG: Heterogeneous Processing-in-Memory Acceleration for Retrieval-augmented Generation**. Chaoqiang Liu, Haifeng Liu, **Dan Chen*,** Yu Huang, Yi Zhang, Wenjing Xiao, Xiaofei Liao, Hai Jin. ACM/IEEE International Symposium on Computer Architecture (ISCA), 2025, pp. 884–898.

3. **RAGNMP: Leveraging Elimination Tree to Accelerate RAG with Near-Memory Processing**. **Dan Chen,** Huize Li, Zhaoying Li, Huiying Lan, Xiang Liu, Zerui Li, Dan Wu, Bin Gao, Tulika Mitra. ACM/IEEE Design Automation Conference (DAC), 2026.

4. **GraphFly: Efficient Asynchronous Streaming Graphs Processing via Dependency-Flow**. **Dan Chen,** Chuangyi Gui, Yi Zhang, Hai Jin, Long Zheng, Yu Huang, Xiaofei Liao. International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2022.

5. **SeIM: In-Memory Acceleration for Approximate Nearest Neighbor Search**. Chaoqiang Liu, **Dan Chen*,** Yu Huang, Wenjing Xiao, Haifeng Liu, Yi Zhang, Huize Li, Xiaofei Liao, Hai Jin. ACM/IEEE Design Automation Conference (DAC), 2025.

6. **Rewire: Advancing CGRA Mapping Through a Consolidated Routing Paradigm**. Zhaoying Li, Dan Wu, Dhananjaya Wijerathne, **Dan Chen*,** Huize Li, Cheng Tan, Tulika Mitra. ACM/IEEE Design Automation Conference (DAC), 2025.

7. **HighP: In-Memory Acceleration of SpGEMM With High Bank-Level Parallelism**. **Dan Chen,** Huize Li, Huiying Lan, Zhaoying Li, Pengcheng Yao, Tulika Mitra. IEEE Transactions on Computers, vol. 75, no. 1, pp. 139–151, 2026.

8. **Accelerating Graph Convolutional Networks Through a PIM-Accelerated Approach**. Hai Jin, **Dan Chen*,** Long Zheng, Yu Huang, Pengcheng Yao, Jin Zhao, Xiaofei Liao, Wenbin Jiang. IEEE Transactions on Computers, vol. 72, no. 9, pp. 2628–2640, 2023.

9. **SADIMM: Accelerating Sparse Attention Using DIMM-Based Near-Memory Processing**. Huize Li, **Dan Chen*,** Tulika Mitra. IEEE Transactions on Computers, vol. 74, no. 2, pp. 542–554, 2025.

10. **A General Offloading Approach for Near-DRAM Processing-In-Memory Architectures**. **Dan Chen,** Hai Jin, Long Zheng, Yu Huang, Pengcheng Yao, Chuangyi Gui, Qinggang Wang, Haifeng Liu, Haiheng He, Xiaofei Liao, Ran Zheng. IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2022, pp. 246–257.

11. **GraphMetaP: Efficient MetaPath Generation for Dynamic Heterogeneous Graph Models**. Haiheng He, **Dan Chen (**co-first author), Long Zheng, Yu Huang, Haifeng Liu, Chaoqiang Liu, Xiaofei Liao, Hai Jin. IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2023, pp. 13–24.

12. **ChainPIM: A ReRAM-Based Processing-in-Memory Accelerator for HGNNs via Chain Structure**. Wenjing Xiao, Jianyu Wang, **Dan Chen*,** Chenglong Shi, Xin Ling, Min Chen, Thomas Wu. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 44, no. 7, pp. 2516–2529, 2025.

13. **SPLIM: Bridging the Gap Between Unstructured SpGEMM and Structured In-Situ Computing**. Huize Li, **Dan Chen*,** Tulika Mitra. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 44, no. 6, pp. 2412–2423, 2025.

14. **HyAtten: Hybrid Photonic-Digital Architecture for Accelerating Attention Mechanism**. Huize Li, **Dan Chen*,** Tulika Mitra. Design, Automation & Test in Europe Conference (DATE), 2025.

Awards

• HGNN accelerator research selected for the **CHIP100 World Annual Chip Contribution List** (2022–2023).

• Meritorious Winner, Interdisciplinary Contest in Modeling (ICM), United States (2017).

Granted Patents

Method and System for Incremental Metapath Storage and Dynamic Maintenance. Long Zheng, Haiheng He, Xiaofei Liao, Hai Jin, **Dan Chen**, and Yu Huang. U.S. Patent US12572594B2, granted 10 March 2026.

Key Research Contributions

1. Green AI Infrastructure

Advancing Green AI through low-cost, energy-efficient AI infrastructure while delivering high performance.

• **HeterRAG:** Proposed a solution combining complementary memory technologies for low-cost, efficient retrieval-augmented generation (RAG). *(ISCA 2025.)*

• **RAGNMP:** Developed a general-purpose near-memory accelerator supporting diverse workloads and evolving RAG algorithms. *(DAC 2026.)*

2. Graph Computing Systems and Accelerators

• **MetaNMP:** Pioneered the first accelerator for heterogeneous graph neural networks, overcoming excessive intermediate storage and redundant computation through hardware–software co-design. This research was selected for the CHIP100 World Annual Chip Contribution List (2022–2023). *(ISCA 2023.)*

• **GraphMetaP:** Co-developed an incremental metapath maintenance method for dynamic heterogeneous graphs, enabling efficient updates without recomputing all metapath instances. Co-inventor of the related granted U.S. patent, US12572594B2. *(IPDPS 2023.)*

3. NMP/PIM Architectures and Computation Offloading

• **Architecture design:** Developed efficient NMP/PIM architectures for AI systems and core computational kernels, addressing data-movement bottlenecks in conventional von Neumann architectures. *(HighP, TC 2026; SPLIM, TCAD 2025; SADIMM, TC 2025.)*

• **Computation offloading:** Proposed computation-offloading techniques to facilitate the integration of NMP/PIM into conventional computing systems. *(IOTPIM, IPDPS 2022; gPIM, TC 2023.)*